MUHAMMAD ABBAS

Graduate Botanist ‖ Gold Medalist ‖ Researcher in Plant Stress Physiology ‖ Scientific Writer

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PERSONAL PROFILE

A highly motivated and academically distinguished Botany graduate (Gold Medalist) with research experience in plant stress physiology, experimental design, and statistical data analysis. Proficient in laboratory techniques, literature review, and scientific writing, with experience contributing to research manuscripts and review articles. Eager to join the Master of Science in Biology at Pittsburg State University, where the research expertise of Professor Daniel Zurek in plant molecular biology, biotechnology, and the development of disease-resistant transgenic crops aligns directly with my academic background and long-term goals in plant stress biology, crop resilience, and sustainable agriculture.

EDUCATION

Bachelor of Science (B.S.) – Botany

Sept 2025

University of Swat, Swat, Pakistan

CGPA: 4.0/4.0 (91.7%), **Rank:** 1st/34

Undergraduate Thesis: *"Effect of Proline Seed Priming on Salt Stress Tolerance of Wild and Cultivated Lentils".*

Supervisor: Dr. Ahmad Ali

Relevant Coursework: Plant Physiology-I and II, Plant Biochemistry-I and II, Plant Ecology-I and II, Molecular Biology, Cell Biology, Genetics & Evolution, Genetics-I and II, Mycology and Plant Pathology, Bacteriology and Virology, Biodiversity and Conservation, Environmental Biology, Research Methodology, and Biostatistics

RESEARCH INTERESTS

- Plant Stress Physiology
- Plant-Microbe Interactions
- Sustainable Agriculture
- Plant Tolerance to Abiotic & Biotic Stresses
- Genetic Engineering for Stress Resilience
- Transgenic Crop Development

RESEARCH EXPERIENCE

Undergraduate Researcher

July 2024 – August 2025

Plant Physiology Laboratory, University of Swat

Supervisor: Dr. Ahmad Ali

- Designed and conducted petri dish and pot experiments to investigate the efficacy of proline priming on the growth and stress tolerance of Lentils (*Lens culinaris* Medik.) under salt stress
- Performed morpho-physiological and biochemical assays, including chlorophyll quantification, proline determination, and antioxidant enzyme activities (SOD and POD) using UV-Vis Spectrophotometry
- Utilized the SmartRoot software (ImageJ) to measure the root system architecture (root length, diameter, surface area, and volume) of lentils under different treatment conditions
- Performed statistical analysis on experimental datasets using MS Excel and analytical software Statistix 8.1
- Collaborated with fellow researchers by assisting in experiments, data analysis, and thesis writing

TEACHING EXPERIENCE

Tutor

Feb 2019 – July 2024

Home Tutor, Balogram, Swat, Pakistan

- Provided personalized one-on-one tutoring, leading to an average 30% improvement in student grades
- Designed personalized learning plans to meet each student's needs, improving their understanding and retention of core subjects, resulting in students achieving positions in their class
- Gained a reputation in the community for effective tutoring and consistent student success

School Teacher Sept 2020 – Oct 2021

Danish Model High School, Balogram, Swat, Pakistan

- Taught Science, Mathematics, and English to Grades 3-7 at an underprivileged school in Swat
- Designed and delivered hands-on science lessons using practical demonstrations and everyday examples to improve student understanding and classroom participation by 40%
- Recognized as list of teachers ranked excellent by students for using effective teaching methods

PROFESSIONAL EXPERIENCE

Scientific Writing Intern Aug 2025 – Nov 2025

Multiomics Academy of Research and Science (MARS), Pakistan

- Learned how to identify research gaps, select a topic for thesis, research article, and reviews (narrative, scoping, systematic, and meta-analysis), create an outline, and select a journal based on scope, indexing, and impact
- Developed skills in advanced literature searching (Google Scholar, PubMed, Scopus, Web of Science) and managing references and citation styles with EndNote and Zotero
- Gained hands-on training in research methodologies and biostatistics, with practical use of SPSS, GraphPad Prism, and R for data analysis and visualization
- **Certificate**: Awarded Certificate of Completion

Research Intern Feb – May 2025

Research Minds **–** NOVA Research Internship Program*, Pakistan*

- Completed a 4-month research internship, building skills in research design and methodology, developed skills in data analysis and visualization using SPSS and R, academic and scientific writing, research proposal and grant writing, literature review, bibliography management, and using AI tools for research
- Co-authored a mini review article in collaboration with a team, contributing to literature synthesis, manuscript drafting, and iterative revision, gaining hands-on experience in collaborative scientific writing and interdisciplinary teamwork
- **Certificate**: Awarded Certificate of Completion

PUBLICATIONS

Thesis

Abbas, M., & Ismail, M. (2025). "*Effect of proline seed priming on salt stress tolerance of wild and cultivated lentils*"**.** Unpublished B.S. research project submitted to Department of Botany, University of Swat, Pakistan

Works in Progress

1. Safdar, A., Afzal, S., Saleem, A., Murtaza, M., Fatima, J., & **Abbas, M**. Comparative genomic and metabolomic analysis of aromatic medicinal plants for the discovery of novel biosynthetic gene clusters related to neurodegenerative diseases. *Phytochemistry Reviews*. (Submitted, July 2026)
2. **Abbas, M.,** Ismail, M., Hussain, S., & Ali, A. Comparative responses of exogenously applied proline on germination and seedling physio-biochemical attributes of *Lens culinaris* under salt stress. (Research Article, Manuscript in Preparation)

Workshops/Conferences/Webinars

- International Conference on Sustainable Forest Management | University of Swat, Pakistan (2025)
- Pannel Discussion on Using Agricultural Innovation to Protect Biodiversity | CGIAR & Crop Trust (2024)
- Plant Cell Focus Issue Webinar on Photosynthesis | American Society of Plant Biologists (ASPB) (2024)
- Climate Change and Phytosanitary Measures Global Webinar Series | IPPC & FAO (2024)

HONORS, AWARDS & SCHOLARSHIPS

- Gold Medalist in BS-Botany for 1st position and CGPA 4.0/4.0, University of Swat, Pakistan (2026)
- Awarded Certificate of Merit for 1st position in BS-Botany, University of Swat, Pakistan (2026)
- Recipient of Merit Scholarship, University of Swat, Pakistan (2021-2025)
- Awarded a Laptop on Merit under Prime Minister's Youth Laptop Scheme, Government of Pakistan (2024)
- PM's Acknowledgment Letter for Environmental Contributions – Green Youth Movement Club (2024)
- Academic Excellence Scholarship – Full School Fee Waiver, Islamic Studies Centre, Balogram (2012-2017)

APPLICABLE SKILLS

Plant Science Laboratory Skills

- **Physiological assays:** Relative Water Content (RWC) determination; Chlorophyll a, b, and carotenoids quantification
- **Biochemical assays:** Proline determination, and antioxidant enzyme activities (SOD and POD)
- Measuring root system architecture using SmartRoot software (ImageJ)

Computational & Statistical Skills

- Microsoft Office packages (Word, Excel, PowerPoint); Google Suit (Google Docs, Sheets, Drive, Forms, Gmail)
- Statistical analysis ability with Microsoft Excel, Statistix 8.1, SPSS, GraphPad Prism, and R
- Reference Management and Bibliographic tools: EndNote and Zotero

Research & Publication

- Excellent ability with searching in advanced scientific databases: Google Scholar, PubMed, Scopus, and WoS
- Excellent research ability with knowledge of different citation and referencing styles, including APA and MPA
- Proficient in managing references and using different citation styles using EndNote/Zotero
- Excellent ability with using AI-tools for academic research

Language Skills

- Pashto, Urdu, and English (Proficient; Oral and Written)

Interpersonal Skills

- Communication skills, Intercultural competency, Critical thinking, Problem-solving, Decision making, Time management, Project management, Teamwork, Leadership

VOLUNTEER EXPERIENCE/EXTRACURRICULAR ACTIVITIES

1. **Green Youth Movement Club, University of Swat** (2022-2025)
   - Participated in plantation, clean-up drives, eco-tourism visits, and poster campaigns on water conservation and mountain restoration
2. **International Mountain Day – Poster Competition** (2023)
   - Designed and presented a poster on "Restoring Mountain Ecosystems", illustrating degraded vs. restored mountain ecosystems; engaged with attendees, interacted with panelists, and received recognition and mountain-themed gifts

REFEREES

Dr. Ahmad Ali
Associate Professor, Department of Botany,
University of Swat
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Relationship – Undergraduate Supervisor

Dr. Junaid Khan
Assistant Professor, Department of Botany,
University of Swat
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Relationship – Undergraduate Teacher